

Question			Solution	Marks
5	(a)		$W_{\text{on}} = P \Delta V$ $= -2.5 \times 10^3 \times (3.0 - 1.0)$ $= -5000 \text{ J (Since gas is expanding, work is done by the gas)}$ $\Delta U = Q_{\text{to}} + W_{\text{on}}$ $= 12500 - 2.5 \times 10^3 \times (3.0 - 1.0)$ $= 7500 \text{ J}$	C1 A1
	(b)	(i)	(little/no volume change so) little/no external work done thermal energy supplied to provide latent heat internal energy increases	B1 M1 A1
		(ii)	no thermal energy enters or leaves the gas (rapid) increase in volume, gas does work against the atmosphere internal energy decreases	B1 M1 A1

Question			Solution	Marks
6	(a)	(i)	Arrow points to the right	B1